Lab: 13

Structures in C++

Objectives

In this lab we will learn:

- To group different types of data under one name.
- To store related information in a single unit.
- To create user-defined data types in C++.
- To make programs more organized and easy to manage.
- To represent real-world entities such as students, employees, books, etc.
- To simplify handling of large amounts of related data.
- To improve code readability and reusability.
- To allow passing multiple related values to functions together.

1: Structures in C++

A structure in C++ is a user-defined data type that allows us to combine different types of data items under a single name. It is used to store related information together in one unit.

- A structure is a collection of simple variables that have same or different data types:
- int, float, char and so on.
- This is unlike the array in which all the variables must be the same type.
- The data items in a structure are called the members of the structure.
- The closing brace in the structure type declaration must be followed by a semicolon.
- It is important to understand that a structure type declaration does not tell the compiler to reserve any space in memory.
- All a structure declaration does is, it defines the‘form’of the structure.
- One structure can be nested within another structure.
- Like an ordinary variable, a structure variable can also be passed to a function.
- We may either pass individual structure elements or the entire structure variable at one go.

2: Defining a Structure

To define a structure, we need to use the keyword struct. The general form of a structure definition in C++ is given as follows

Syntax:

```
struct StructureName

{

dataType member1;

dataType member2;

dataType member3;

};
```

Example: **Structure with one structure variable**

```
#include
using namespace std;
struct part:            // declare a structure
{
  int modelnumber;      // ID number of car (structure member)
  int partnumber;       // ID number of car part (structure member)
   float cost;          // cost of part (structure member)
 };
int main()
{
   part part1;          // define a structure variable
// give values to structure members
   part1.modelnumber = 6244;
   part1.partnumber = 373;
   part1.cost = 217.55F;
// display structure members
  cout << " Model : " << part1.modelnumber << endl;
  cout << " Part : " << part1.partnumber << endl;
  cout << " Cost : " << part1.cost << " $" << endl;
 return 0;
}'
```

Output:

Model: 6244

Part: 373

Cost: 217.55 $

Example: Structure with two structure variable

```
#include
using namespace std;
struct part {              // declare a structure
     int modelnumber;    // ID number of car (structure member)
    int partnumber;        // ID number of car part (member)
    float cost;    };
            // cost of part (structure member)
int main() {
    part part1 = {6244, 373, 217.55F};   // initialize variable
   part part2;                                      //declare a structure variable
// display structure members related to first structure variable,
  cout << " Model : " << part1.modelnumber << endl; 1
  cout << " Part : " << part1.partnumber << endl; 1
  cout << " Cost : " << part1.cost << " $" << endl;
  part2 = part1;                      // assign 1st variable to 2nd (copying)
// display members related to 2nd structure variable, part2
   cout << " Model : " << part2.modelnumber << endl;
   cout << " Part : " << part2.partnumber << endl;
   cout << " Cost : " << part2.cost << " $" << endl;
   return 0; }
```

Output:

Model: 6244

Part: 373

Cost: 217.55 $

Model: 6244

Part: 373

Cost: 217.55 $

Example: Employee Information using Structure

```
#include
using namespace std;
struct Person
{
   char name[50];
   int age;
   float salary;
};
int main() {
  Person p1;
  cout << "Enter Full name: ";
  cin.get(p1.name, 50);
  cout << "Enter age: ";
  cin >> p1.age;
  cout << "Enter salary: ";
```

```
cin >> p1.salary;
 cout << "\nDisplaying Information." << endl;
  cout << "Name: " << p1.name << endl;
  cout << "Age: " << p1.age << endl;
  cout << "Salary: " << p1.salary;
 return 0;}
```

Output:

Enter Full name: Murad Ali

Enter age: 34

Enter salary: 1200

Displaying Information.

Name: Murad Ali

Age: 34

Salary: 1200

Lab Tasks:

Task 1: Structure Declaration and Initialization

- Declare a structure called Person with the following members:
  - name (string)
  - age (integer)
  - address (string).
- Declare a variable called person1 of type Person and initialize its members with values of your choice. Display the values of the members of person1 using dot notation.

```
#include <iostream>        // for input/output
using namespace std;       // to use cout without std::
struct Person              // structure declaration
{
    string name;           // member to store name
    int age;               // member to store age
    string address;        // member to store address
int main(){
    Person person1;                  // creating structure variable
    person1.name = "Asad";        // assigning name
    person1.age = 20;                // assigning age
    person1.address = "Peshawar";            // assigning address
    cout << "Name: " << person1.name ;       // display name
    cout << "Age: " << person1.age ;            // display age
    cout << "Address: " << person1.address ;  // display address
    return 0;   }                                          // program ends
```

Task 2: Structure Declaration and Initialization

- Declare a structure called Student with the following members:
  - name (string)– age (integer)
  - grade (char)
- Declare a variable of type Student called student1 and initialize its members with values of your choice. Display values of student1 structure members.

```
#include <iostream>
using namespace std;
struct Student          // structure declaration
{
    string name;        // student name
    int age;            // student age
    char grade;         // student grade
};
int main()
{
    Student student1;                          // structure variable
    student1.name = "Ali";                     // assigning name
    student1.age = 19;                         // assigning age
    student1.grade = 'A';                      // assigning grade
    cout << "Name: " << student1.name << endl;     // display name
    cout << "Age: " << student1.age << endl;       // display age
    cout << "Grade: " << student1.grade << endl;   // display grade
    return 0;}
```

Task 3: Accessing Structure Elements

- Declare a structure called Book with the following members:
  - title (string)
  - author (string)
  - price (float)
  - pages (integer)
- Declare a variables of type Book called book1 and book2, and initialize its members with values of your choice.
- Display values of structure members of variables book1 and book2 .

```
#include <iostream>

using namespace std;

struct Book                          // structure declaration

{

    string title;                    // book title

    string author;                   // author name

    float price;                     // book price

    int pages;                       // total pages

};

int main()

{

    Book book1;                      // first book variable

    Book book2;                      // second book variable
```

```
book1.title = "C++ Basics";       // title of first book
  book1.author = "Bjarne";           // author
   book1.price = 500.5;               // price
   book1.pages = 300;                 // pages
   book2.title = "Programming";    // title of second book
   book2.author = "John";           // author
   book2.price = 650;                 // price
   book2.pages = 450;                // pages
   cout << "Book 1" << endl;
   cout << book1.title << endl;
   cout << book1.author << endl;
   cout << book1.price << endl;
   cout << book1.pages << endl;
   cout << "Book 2" << endl;
   cout << book2.title << endl;
   cout << book2.author << endl;
   cout << book2.price << endl;
   cout << book2.pages << endl;
   return 0;           // end
}
```